📅 25-Day Java Study Plan (Beginner → Interview Ready)

🔹 Week 1: Java Fundamentals (Days 1–7)

Day 1 – Java Basics

- What is Java? Features
- JVM vs JRE vs JDK ⭐
- Java program structure
- First Java program
- Compilation & execution flow

📝 Practice:

- Print patterns
- Simple input/output programs

Day 2 – Data Types & Operators

- Primitive vs Non-primitive
- Variables & scope
- Type casting
- Operators (all types)

📝 Practice:

- Swap two numbers
- Type conversion programs

Day 3 – Control Statements

- if–else
- switch
- for, while, do-while
- break & continue

📝 Practice:

- Prime number
- Factorial
- Fibonacci series

Day 4 – Arrays

- 1D & 2D arrays
- Array vs ArrayList
- Sorting & searching

📝 Practice:

- Find max/min in array
- Reverse array
- Duplicate elements

Day 5 – Strings

- String vs StringBuilder vs StringBuffer ⭐
- String immutability
- == vs equals()
- Important String methods

📝 Practice:

- Reverse string
- Palindrome string
- Count characters

Day 6 – OOPs (Part 1) ⭐ ⭐ ⭐

- Class & Object
- Constructor
- this keyword
- Encapsulation

📝 Practice:

- Create real-life OOP examples

Day 7 – OOPs (Part 2) ⭐ ⭐ ⭐

- Inheritance
- Polymorphism
- Abstraction

- Interface vs Abstract class

📝 Practice:

- Method overriding examples

🔹 **Week 2: Core Java Concepts (Days 8–14)**

Day 8 – Exception Handling

- Checked vs Unchecked
- try-catch-finally
- throw vs throws
- Custom exceptions

📝 Practice:

- Custom exception program

Day 9 – Wrapper Classes

- Integer, Double, Character
- Autoboxing & Unboxing

📝 Practice:

- Conversion programs

Day 10 – Collections (Part 1) ⭐⭐⭐⭐

- Collection Framework
- List: ArrayList, LinkedList
- Iterator vs ListIterator

📝 Practice:

- Store & iterate objects

Day 11 – Collections (Part 2) ⭐⭐⭐⭐

- Set: HashSet, TreeSet
- Map: HashMap, Hashtable, TreeMap
- equals() & hashCode()

- Internal working of HashMap ⭐

📝 Practice:

- Frequency count using HashMap

Day 12 – Multithreading (Part 1) ⭐⭐⭐⭐

- Thread vs Runnable
- Thread lifecycle
- Creating threads

📝 Practice:

- Basic multithread program

Day 13 – Multithreading (Part 2) ⭐⭐⭐⭐

- Synchronization
- Deadlock
- Volatile
- Inter-thread communication

📝 Practice:

- Synchronized block example

Day 14 – Java Memory Management

- Stack vs Heap ⭐
- Garbage Collection
- finalize()
- Memory leaks

📝 Practice:

- GC concept questions

🔹 **Week 3: Modern Java + Revision (Days 15–25)**

Day 15 – Java 8 Features (Part 1) ⭐⭐⭐⭐

- Lambda expressions

- Functional interfaces

📝 Practice:

- Lambda with Comparator

Day 16 – Java 8 Features (Part 2) ⭐⭐⭐⭐

- Stream API
- map(), filter(), reduce()
- forEach()

📝 Practice:

- Stream programs on list

Day 17 – Optional & Method Reference

- Optional class
- Method references

Day 18 – File Handling

- File class
- Reader & Writer
- Serialization

Day 19 – Important Keywords

- static
- final
- synchronized
- transient
- volatile

Day 20 – Important Java Programs

- Armstrong number
- Palindrome
- Prime number

- Reverse number/string

Day 21 – JVM Deep Dive

- ClassLoader
- Memory areas
- Execution engine

Day 22 – Interview Questions (Theory)

- OOPs questions
- Collections questions
- Multithreading questions

Day 23 – Coding Practice

- Arrays & Strings problems
- HashMap problems
- Thread programs

Day 24 – Mock Interview Day

- Explain OOP concepts loudly
- Explain HashMap internally
- Write code without IDE

Day 25 – Final Revision

- Weak topics
- One-line answers
- Short notes revision